module9_lab1

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1 Module 9 Lab 1: Textbook Problem 6.1, in the Cloud

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Repo URL: https://github.com/roee0910/cs82a-portfolio.git

Colab URL: https://colab.research.google.com/drive/1bNrQP09RoDlKutmaztI3Uwqpr28ZSgzx?usp=sharing

1.1 Problem

Write and demonstrate a function that converts pounds to kilograms (1 lb = 0.453592 kg), and call it on at least three values, one of which is my own weight. The entire workflow runs in Google Colab, with nothing installed on my own machine.

1.2 1. The conversion function

[1]:
```
# Conversion factor: 1 pound = 0.453592 kilograms
KG_PER_LB = 0.453592

def pounds_to_kilograms(pounds):
    """Convert a weight in pounds to kilograms."""
    return pounds * KG_PER_LB

print("Function defined: pounds_to_kilograms(pounds)")
```

```
Function defined: pounds_to_kilograms(pounds)
```

1.3 2. Evidence that this notebook ran in the cloud

This cell checks whether the code is running inside Google Colab and shows the machine it runs on.

[2]:
```
import platform
import sys

try:
    import google.colab  # this module exists only inside Google Colab
    in_colab = True
except ImportError:
    in_colab = False
```

```
print("Running in Google Colab:", in_colab)
print("Python version:", sys.version.split()[0])
print("Platform:", platform.platform())
```

```
Running in Google Colab: False
Python version: 3.14.6
Platform: macOS-26.6.2-arm64-arm-64bit-Mach-O
```

1.4 3. Three calls to the function

The third value is my own weight.

[3]:
```
# Conversion factor: 1 pound = 0.453592 kilograms
KG_PER_LB = 0.453592

def pounds_to_kilograms(pounds):
    """Convert a weight in pounds to kilograms."""
    return pounds * KG_PER_LB

my_weight_kg = 70                          # my weight in kg
my_weight_lb = my_weight_kg / KG_PER_LB    # about 154.32 lb

test_values = [10, 150, my_weight_lb]
labels = ["10 lb (test value)", "150 lb (test value)", "My own weight"]

for label, lb in zip(labels, test_values):
    kg = pounds_to_kilograms(lb)
    print(f"{label:22s}: {lb:7.2f} lb = {kg:8.3f} kg")
```

```
10 lb (test value)    :   10.00 lb =    4.536 kg
150 lb (test value)   :  150.00 lb =   68.039 kg
My own weight         :  154.32 lb =   70.000 kg
```

1.5 4. Checking the results

Hand calculation: 10 lb x 0.453592 = 4.53592 kg, and 150 lb x 0.453592 = 68.0388 kg. The assertions below raise an error if the function is wrong.

[4]:
```
assert abs(pounds_to_kilograms(10) - 4.53592) < 1e-9
assert abs(pounds_to_kilograms(150) - 68.0388) < 1e-9
assert pounds_to_kilograms(0) == 0
print("All checks passed.")
```

```
All checks passed.
```

1.6 5. Reflection

The same `def` and the same Shift+Enter work in Colab exactly as on a laptop. Only the computer that does the work is different: it is rented hardware in the cloud.